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The crypto conundrum: Is it right for you?

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WS Global CIO Office



Although numerous cryptocurrencies exist, very few have significant market scale. The few that are actively traded generate daily trading volumes exceeding USD 100bn. This indicates digital currencies are rapidly transitioning to become mainstream investment assets.

This market's nascency could promise material upside. However, investors should stay mindful of extreme volatility, cyber risks, regulatory uncertainty and environmental impact.

This report offers a comprehensive evaluation of cryptocurrency as an investment option, highlighting both its risks and rewards.

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Understanding the landscape

Cryptocurrency is a digital or virtual form of money secured by cryptography and typically operating on blockchain networks. It allows peer-to-peer value transfer without reliance on banks or central authorities, underpinning its decentralised nature.

The global cryptocurrency industry was valued at approximately USD 3tn in 2025 according to CoinGecko 2025 Annual Crypto Industry report. The crypto market has historically generated strong returns and provided meaningful diversification, largely owing to its low or negative correlation with traditional assets.

Fig. 1

Bitcoin vs other assets

Returns, volatility and max drawdown

Assets	CAGR	Annualised volatility	Max drawdown
Bitcoin	3.81%	56.45%	-76.63%
Ethereum	1.97%	76.33%	-79.35%
S&P500	24.30%	17.38%	-20.43%
US Bonds (TLT)	-5.87%	15.87%	-43.70%
Gold	11.60%	16.78%	-25.43%

Bitcoin correlation with other assets over different timeframes



Source: Standard Chartered



The combination of blockchain technology, cryptography and a decentralised network creates a financial ecosystem that enables secure and transparent transactions in digital assets. Understanding the asset's nuances is, therefore, important before venturing into this investment space.

Mining

Several cryptocurrencies are created by a process called mining, wherein crypto 'miners' are rewarded with a crypto coin (or a share of the coin) when they solve complex mathematical problems, typically on high-performance computers. These coins can then be traded on a digital exchange.

Staking

This process entails locking up crypto holdings to support blockchain operations such as proof-of-stake consensus. Investors who stake are compensated for these efforts with additional tokens, making it a good source of passive income. Staking helps secure the network and validate transactions. Risks include penalties for misconduct on the network and devaluation of tokens.

Initial coin offerings (ICOs)

A method of raising capital for blockchain projects by selling cryptocurrency tokens to investors, functioning similarly to an IPO. Investors can buy cryptocurrency tokens before its launch, often at discounted prices. However, risks are significantly high due to the lack of regulation.

Digital exchanges

Cryptocurrencies can be bought and sold on online platforms that are broadly categorised as centralised exchanges (CEXs), such as Coinbase and Binance, and decentralised exchanges (DEXs), such as PancakeSwap and Uniswap. CEXs are operated by a company that controls the platform, manages the order books, holds user funds in custody and oversees operational nuances. DEXs run on smart contracts, do not hold user funds and allow peer-to-peer trading without any intermediaries. As of today, there are over 200 active crypto exchanges worldwide.

Buying and selling

Trading in crypto occurs in pairs; each pair has a base currency and a quote currency. Pair trading allows for transactions between two cryptocurrencies, bypassing fiat currencies. For example, BTC/ETH trading at 15 represents the trading of 1 Bitcoin against 15 Ethereum. However, crypto trading is not restricted to crypto only pairs; several trades involve fiat or stablecoins as the quote currency.

Consensus protocol

This is a method by which decentralised network participants (nodes) validate transactions. This process ensures that all transactions are secure and verified, preventing fraud and double-spending while maintaining the blockchain's integrity.

Fig. 2

Types of consensus protocols

Aspect	Proof of work (PoW)	Proof of stake (PoS)	Delegated proof of stake (DPoS)	Proof of authority (PoA)	Practical byzantine fault tolerance (PBFT)
Method	Miners solve cryptographic problems to validate transactions	Validators are chosen based on quantity of assets staked	Delegates are elected to perform transaction validation duties	Validators are determined by their reputation and established trust	Consensus is achieved by communication with pre-determined validators
Energy efficiency	Low	High	Very high	Very high	Very high
Security	High	Medium/High	Low	High	High
Decentralisation	Low (requires powerful systems)	High (decentralised validators)	Moderate (voters control delegates)	Low (centralised)	Moderate (centralised)
Transaction speed	Slow (mining time)	Fast	Very fast (limited validators)	Very fast (limited validators)	Fast (designed to scale)
Examples	Bitcoin, Litecoin	Ethereum	EOS	VeChain, Ethereum (private chain)	Zilliqa (hybrid), Cosmos/Tendermint

Source: Standard Chartered

Digital wallets

These wallets store cryptos securely and facilitate transactions. They consist of private and public keys. Broadly, wallets are categorised as follows:

- **Hot wallets** are connected to the internet and are easy to access for frequent trading. MetaMask and Trust wallet are widely used hot wallets.
- **Cold wallets** store assets offline and offer better security. Ledger and Trezor are popular cold wallets.

Certain wallets also offer portfolio management tools to track and analyse holdings and allow access to decentralised apps (DApps). For example, MetaMask can interact with a DEX (e.g., Uniswap).

It is important to be aware of the legal, tax and regulatory implications of cryptocurrencies in the country where investments are made, as these can vary significantly across jurisdictions.

Global regulations

The evolving nature of regulations for cryptocurrencies across economies reflects the potential of emerging technologies and risks associated with such digital assets. There is growing acceptance of cryptocurrencies in both Developed and Emerging Markets.

For instance, after the US approved spot Bitcoin ETFs, Hong Kong quickly followed by launching spot Bitcoin and Ether ETFs in 2024. There is growing interest among investors to gain exposure to crypto assets. Around 39% of Singaporeans (over 1,500 surveyed) indicated their interest in investing in cryptocurrencies. Growing interest and increased market participation have reduced regulatory uncertainty around Bitcoin, encouraging broader cryptocurrency adoption. Several countries such as Switzerland, the UAE, Hong Kong, South Korea and Singapore are known for being crypto-friendly and are increasingly positioning themselves as being potential hubs for the development of digital assets, including cryptocurrency and blockchain technology.

Here are some of the key aspects that investors should evaluate in global regulations before investing:

Even though digital assets run on decentralised networks, investor safeguards such as fraud prevention, disclosure rules and KYC/AML checks depend entirely on local laws.



Investors should assess whether a jurisdiction enforces strong guardrails and oversight.

Some jurisdictions adopt comprehensive virtual-asset frameworks with strict licensing and reporting rules for exchanges, while others have minimal or fragmented oversight.



Investors could benefit from investing in jurisdictions that have robust digital asset frameworks.

Cryptocurrencies, although globally accessible, are susceptible to cross-border inconsistencies. An exchange regulated in one jurisdiction may offer little protection to investors in another.



Investors need to confirm whether regulatory coverage extends to their residency and the platforms they use.

Investment options

Mining was the earliest form of investing in cryptocurrencies; however, much has changed over the past few years. Investing through mining is not everyone's cup of tea. It requires knowledge to solve complex algorithmic problems and hardware with high computational capacity.

Given these limitations and the outsized historical returns of this asset class, many investment options have emerged in this space, with varying degrees of risk, complexity and return potential. Let's look at some popular methods of investing in cryptocurrencies.

Direct purchase of cryptocurrencies

Direct purchase of cryptocurrencies is a straightforward method to invest in cryptocurrencies. The investor purchases digital currencies such as Bitcoin or Ethereum via exchanges such as Binance and Coinbase. The purchase can be done with fiat currency or stablecoins (tracks traditional currencies). The cryptocurrencies bought are stored securely in hot or cold wallets. While this approach offers full ownership and flexibility, it requires caution as the risks are quite high. It is ideal for investors who actively manage their investments. However, thorough research and a sound understanding of risks are essential while dealing with these instruments.

Cryptocurrency funds

Cryptocurrency funds are a passive way to gain exposure to this asset class. Crypto funds pool money and invest in a range of digital assets including cryptocurrencies. They offer liquidity, diversification and professional management. While the risk remains high, it is slightly lower than that of direct investing.

- **Crypto ETFs** track the performance of a single or basket of digital assets or cryptocurrencies. Several countries, including the US, Hong Kong and South Korea, allow investments via spot Bitcoin or broader crypto ETFs. This is a significant milestone as it offers a regulated alternative to direct investing, which could drive a wider acceptance of cryptos among investors.
- **Crypto index funds** replicate a crypto index that invests in multiple cryptocurrencies (typically top 10 or 20 cryptos based on market capitalisation). The returns track the index. It is akin to index equity funds, except that the underlying assets are digital. This is yet another passive way of investing in digital assets and carries relatively slightly lower risk because of its inherent diversification.
- **Crypto mutual funds** invest in a basket of cryptocurrencies or blockchain-related assets. They are similar to traditional mutual funds, except that they have digital assets in the portfolio. They are professionally and actively managed. The cost is usually higher than ETFs and index funds.
- **Crypto hedge funds** are specialised investment options that are actively managed by employing complex strategies to generate outsized returns. Some sophisticated strategies include statistical arbitrage, event-driven positioning, quantitative and algorithmic trading. These investments are available for accredited investors who meet the requirements of minimum net worth or income.

- **Crypto venture capital (VC)** are options that focus on investing in early stage blockchain projects and cryptocurrency start-ups. **Crypto private equity funds** target more mature crypto firms or projects that require capital for expanding their business. The rate of failure is quite high in early startups, making crypto VCs high-risk instruments. Crypto private equity funds are also high risk but less risky than VC stage investing. These funds typically have an investment horizon of 5-10 years. Investors gain equity or tokens in exchange for capital invested.

Crypto lending

Crypto lending refers to peer-to-peer lending of digital assets in exchange for interest. Broadly, crypto lending platforms are categorised as centralised (e.g., BlockFi, Nexo) or decentralised (e.g. Aave or Compound). While centralised lending platforms facilitate loan and hold collateral, decentralised platforms allow direct lending between peers. Interest rates are determined by the risk level and collateral provided. These transactions are exposed to market volatility, platform insolvency and counterparty default.

Yield farming

Yield farming refers to providing liquidity to decentralised platforms such as Uniswap or Yearn Finance to earn tokens or fees in return. Yield farmers provide cryptocurrency to liquidity pools, facilitating trading on the platform. They are similar to market makers in traditional financial markets. While this mode of crypto exposure can provide high returns, there are risks such as smart contract vulnerabilities and loss of asset value in the liquidity pool.



Key risks

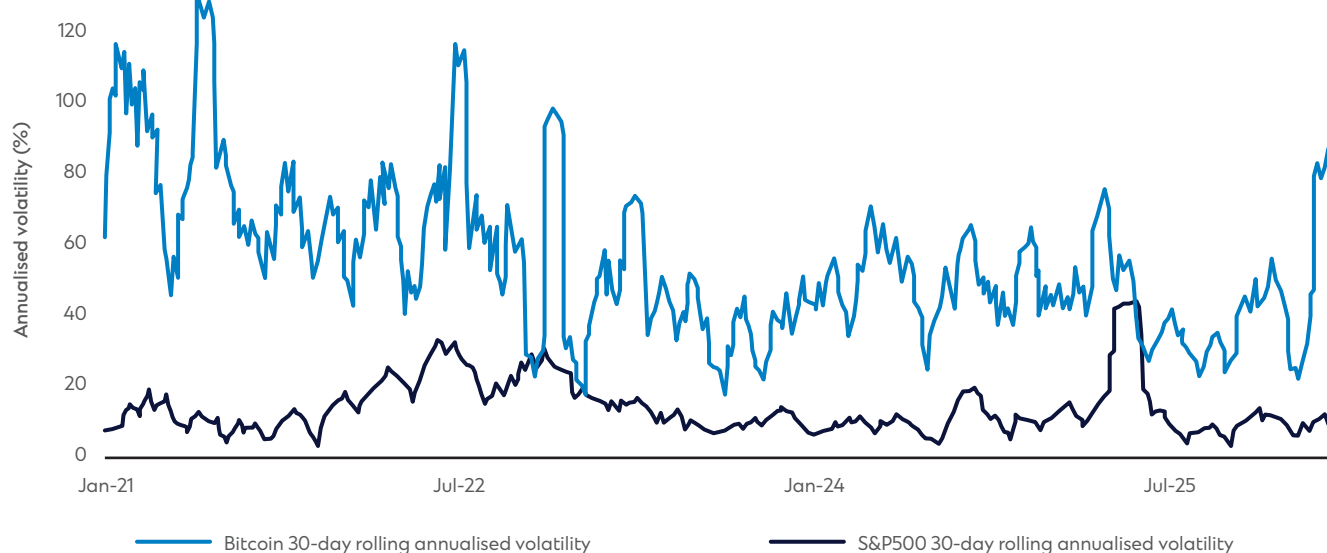
Cryptocurrencies are not for everyone as they carry several inherent and unique risks.

High volatility

It is a well-known fact that cryptocurrencies are a highly volatile investment option, both on an absolute basis and when juxtaposed alongside traditional assets. For example, Amplify ETF's report found that bitcoin's volatility was nearly 4x the average volatility of equities.

Fig. 3

Volatility of Bitcoin vs other assets



Source: Standard Chartered

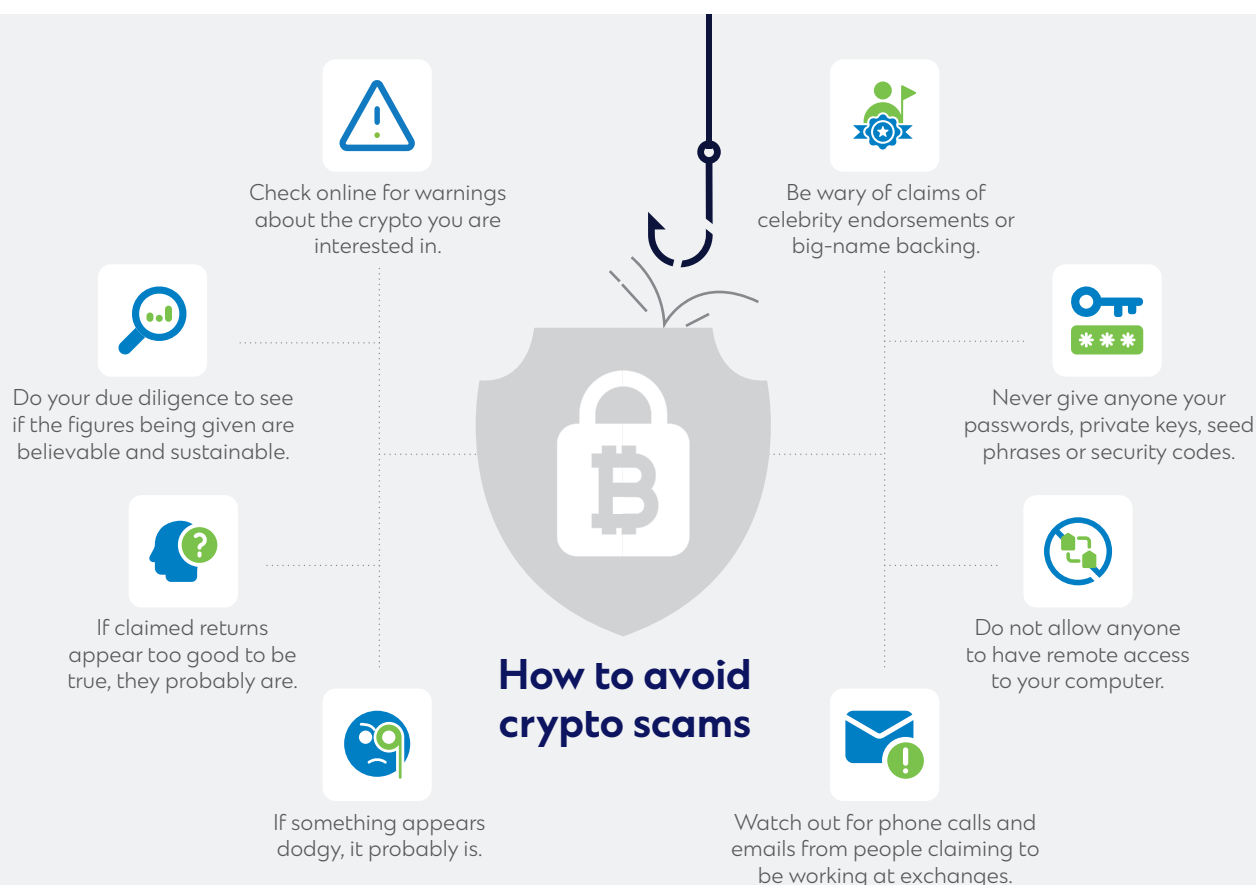
That said, high risk may translate to high return. This is evident from Bitcoin's Sharpe ratio (a measure of return earned per unit of risk taken) of 0.96 for the period between 2020 and 2024 compared with S&P 500's Sharpe ratio of 0.65, indicating a higher risk-adjusted return. However, it is important to highlight that cryptocurrencies are subject to massive drawdowns. For example, in 2022, Bitcoin's price dropped by 77% from its peak. This fall was attributed to macroeconomic conditions, regulatory uncertainty and general drawdown in the crypto markets.

Regulatory uncertainty

The crypto landscape is rapidly changing, posing a challenge to both governments and investors. The lack of standardised frameworks and differing approaches across jurisdictions do not offer a conducive environment for building trust among market participants. Trust is crucial for increased investor participation, market efficiency and contract efficacy.

Cyber susceptibility

A Statista report states that hacks have been the predominant driver of crypto losses since 2022. The year 2022 was a particularly significant year when total crypto-related crime stood at a record USD 39.6bn. It is important that investors stay vigilant of their exposures and take steps to avoid being scammed.



Source: Standard Chartered

Private key vulnerability

The vulnerability of private keys in cryptocurrencies is a significant risk. There have been many incidents of investors forgetting private keys, leading to a complete loss of access to one's cryptocurrency holdings. Around 20% of Bitcoins in circulation have been considered lost, primarily due to forgotten private keys and seed phrases (words that recover your crypto wallet). The recovery options for lost private keys are limited. Reports estimate that only about 2.5% of the lost coins may be recoverable. Stefan Thomas, German programmer and crypto investor, lost access to over 7,000 Bitcoins stored on an encrypted USB drive due to his inability to recall the password required to unlock the device. Several investors under-estimate the importance of safeguarding private keys and seed phrases.



Liquidity risks

The crypto universe's unique nature and inherent volatility have at times led to extreme liquidity risks. For example, in 2022, when Bitcoin dipped sharply, most exchanges experienced unprecedented trading volumes, and this resulted in a liquidity crisis. A liquidity crisis is characterised by large bid-ask spreads and delayed transactions.

Lack of investor protection protocol

The lack of a statutory framework in the digital assets space leaves investors exposed to frauds, scams, phishing attacks and data breaches. Many governing authorities including the Monetary Authority of Singapore (MAS) have issued warnings highlighting the risks investors face owing to inadequate protection against crypto scams, fraud and loss of digital assets.

Emerging trends

The below are some emerging trends in the crypto world that are shaping the future of digital finance.



Meme coins – a unique category of cryptocurrencies that originated from internet humour – continues to gain traction against the backdrop of community and social media engagement.

The DeFi sector continues to grow with new tokens being introduced that offer various utilities including staking and yield farming.

Central Bank Digital Currencies (CBDCs) are being considered by many jurisdictions; this could revolutionise the financial system. These can help support faster transactions across borders at minimal cost while also ensuring optimal security. Unlike cryptocurrencies, these currencies will have the ability to integrate with existing financial ecosystems.

Green cryptos are quickly rising in popularity as investors prioritise sustainable options even within the digital asset space. These cryptos follow ESG standards and track their carbon footprint. Projects such as Chia use lower energy to validate transactions.



Conclusion

This publication provides foundational information to help investors navigate the crypto world responsibly. While the outsized returns potential of cryptocurrencies may be tempting, it is important to tread cautiously due to the unique risks involved, such as cyber vulnerability and regulatory shifts. Over the next few years, regulatory frameworks for digital assets are likely to become more structured and standardised. This will not only bring better clarity but also strengthen investor confidence.

Investors should approach cryptocurrencies with an opportunistic and cautious mindset. That said, investors can explore this opportunity provided it aligns with their risk profile and broader wealth goals.

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